Janit Berwal

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Education

Guru Gobind Singh Indraprastha University Aug 2023 – Jul 2027
Bachelor of Technology, Information Technology | **CGPA: 9/10** New Delhi, India

Technical Skills

Languages & Core CS: C, C++, Python, JavaScript, SQL, Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), System Design, DBMS

Frontend Technologies: HTML, CSS, Tailwind, Bootstrap, React, Next.js

Backend & Database: NodeJS, ExpressJS, FastAPI, REST APIs, Socket.io, MySQL, PostgreSQL, Prisma ORM

Cloud, DevOps & Tools: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Redis, MinIO, Git, Postman

Machine Learning & Testing: CNNs, YOLOv8, OpenCV, Pandas, NumPy, Manual/API Testing, Test Case Design

Experience

Developer, Student Achievement Management System (SAMS) *Jun 2026 – Present*
GGSIP University

- Architected a **three-tier role-based system** (Student, Mentor, HOD) using **Next.js** and **Tailwind CSS**, scaling to support **500+ students annually** while managing a **1:30 mentor-to-student allocation ratio**.
- Designed a **verification workflow** integrated with **Cloudinary** for secure proof document uploads, routing hundreds of extracurricular, co-curricular, project, and research achievement records to mentors each semester.
- Built **HOD-level admin controls** for mentor assignment/reassignment, approval overrides, and multi-level (student/class/batch-wise) Excel reporting, eliminating manual administrative tracking.
- Engineered the backend database schema using **Prisma ORM** and **PostgreSQL** to support approval audit trails and strict role-scoped access control.

Full Stack Web Developer Intern *Jan 2026 – Feb 2026*
Accounts Bureau

- Engineered a Full-Stack OCR Pipeline utilizing **LLMs (Gemini 1.5 Flash)** and **OpenCV** to automate data extraction from physical business cards, achieving **80-90% accuracy** into structured digital formats.
- Implemented Advanced Regex & NLP logic for data validation, executing comprehensive **Software Testing** to ensure formatting consistency across simulated contact entries.
- Architected a **FastAPI Backend** with **REST APIs**, integrated with a fully normalized **MySQL RDBMS** for persistent storage, improving query efficiency and data integrity over flat-file exports.

Projects

BhejaFry – Real-Time Multiplayer Trivia Game | *React, Node.js, Socket.io, PostgreSQL, Redis* *bhejafry.fun*

- Engineered real-time bidirectional communication with **Socket.io** to manage live game rooms and synchronize state across concurrent players with under 200ms latency for up to 50 concurrent users.
- Applied core **System Design** principles to architect a scalable backend using **Redis** for caching and **MinIO** for object storage, deployed on **AWS EC2** to ensure reliability and graceful disconnection handling.

Vehicle Speed Detection System | *Python, YOLOv8, OpenCV, NumPy, Pandas* Github Link

- Built a vehicle speed estimation pipeline using **YOLOv8** object detection, tracking vehicle displacement across frames and calibrating pixel-to-real-world distance to compute speed.
- Processed recorded traffic footage to track multiple vehicles simultaneously across frames, overlaying real-time bounding boxes and computed speed for each tracked vehicle.

Achievements

- Solved more than **550** Data Structures & Algorithms problems on **LeetCode, CodeChef,** and **Codeforces**, strengthening **problem-solving, algorithmic thinking, and analytical skills**.
- **SIH 2024** - Advanced past the Internal College Round, competing against 200+ teams.